

Executive Summary: NYC Yellow Taxi Fare Prediction Project

Progress update for teammates and stakeholders • Prepared by the Data Analytics Team

Summary of Tasks

This phase of the project focused on preparing the NYC Taxi & Limousine Commission trip dataset for predictive modeling and building an initial model to estimate total fare amount. Work completed included:

- Data loading & inspection:** Loaded and inspected the raw trip-record dataset (22,699 records) to understand structure, data types, and quality issues.
- Data cleaning & wrangling:** Converted pickup/drop-off timestamps to datetime format and engineered a trip duration_minutes feature.
- Outlier and anomaly removal:** Identified and removed records with negative durations or fares, zero-distance trips, and total_amount outliers (e.g., \$0 or over \$500), reducing the dataset to 22,535 quality-controlled records.
- Feature engineering:** Extracted pickup_hour, pickup_day_of_week, and pickup_month from timestamp data to capture time-based fare patterns.
- Model development:** Trained and evaluated a Linear Regression model to predict total_amount using 12 trip-level features.

Data Variable Assessment Results

Twelve variables were assessed and retained as predictors of total_amount, combining trip-level, temporal, and categorical features:

- Numerical features (6):** trip_distance, passenger_count, duration_minutes, pickup_hour, pickup_day_of_week, pickup_month.
- Categorical features (6):** VendorID, RatecodeID, PULocationID, DOLocationID, payment_type, store_and_fwd_flag — encoded via one-hot encoding.

Data quality: Cleaning removed roughly 0.7% of records (164 rows) as invalid or extreme outliers, indicating the raw dataset was largely reliable once obvious anomalies were filtered out.

The model's predicted fares closely tracked actual fares across the price range, confirming trip_distance and duration_minutes as the strongest predictors of total_amount.

Recommended Next Steps

- Explore advanced models:** Test Gradient Boosting and Random Forest regressors against the current linear baseline to capture non-linear fare patterns and potentially improve accuracy.
- Run feature importance analysis:** Quantify which predictors (e.g., distance vs. time-of-day) drive fare estimates, to inform pricing and fraud-detection use cases.
- Optimize hyperparameters:** Tune model parameters (e.g., regularization for Ridge/Lasso variants) to reduce error further and guard against overfitting on new data.
- Plan deployment:** Scope integration of the model into a real-time fare estimation workflow, including data pipeline and monitoring requirements.

Bottom line: The initial Linear Regression model already explains 90% of fare variance with an average error under \$2.50 — a strong foundation. Next steps focus on refinement and deployment readiness rather than starting over.